H-B-1051A
January 15, 1973
SUPERSEDING
Int. Fed. Spec. H-B-001051(GSA-FSS)
October 29, 1970
Fed. Spec. H-B-173(IN PART)
October 8, 1948 (see 6.2)

FEDERAL SPECIFICATION

BRUSH, TEST TUBE AND BRUSH, PIPET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for brushes intended for use in cleaning laboratory test tubes and pipets.

1.2 Classification.

1.2.1 Types, styles, classes and sizes. Brushes covered by this specification shall be of the following types, styles, classes and sizes.

Types I - Test tube brush.
  Style A - Dome tip.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Sizes - 1/2 inch, 3/4 inch, 1-1/4 inch, 1-1/2 inch.

  Style B - Tuft tip.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Sizes - 1/2 inch, 3/4 inch, 1 inch, 1-1/8 inch, 1-1/4 inch, 1-3/8 inch, and 1-1/2 inch.

  Style C - Single fan tip.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filaments.
      Size - 1/2 inch.

  Style D - Sponge tip.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Size - 1-1/2 inch.

  Style E - Conical shape w/o tuft.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Size - 1/2 - 13/16 inch (tapered).

  Style F - Conical shape with tuft tip
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Sizes - 3/4 - 1-1/2 inches; 1 - 1-3/4 inches (tapered).

Type II - Pipet brush.
  Style A - Conical.
    Class 1 Hog bristle.
    Class 2 - Synthetic filament.
      Size - 1/4 - 3/4 inch (tapered).

  Style B - Open-end tube.
    Class 1 - Hog bristle.
    Class 2 - Synthetic filament.
      Size - 1/2 inch.

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-S-951 - Synthetic Filaments for Brushes; General Specification for.
R-N-91 - Napthalene, Technical.
PPP-B-1729 - Brushes, Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Bristle. The bristle for the class 1 brushes shall be unbleached, stiff, natural color, resilient hog bristles of uniform length, and shall be cleaned and sterilized as specified in 3.3. The bristles shall be evenly trimmed and shall not include flagged ends.

3.1.2 Synthetic filament. The filament for the class 2 brushes shall be made of synthetic-polymeric amide, meeting the requirements of class N (nylon), type II of H-S-951. The color shall be black, unless otherwise specified (see 6.2).

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3.1.3 Wire. The wire shall be zinc-coated (galvanized) steel wire or corrosion-resistant steel wire normally used for twisted wire brushes. The wire diameter shall be as specified in the tables of dimensional requirements.

3.2 Construction. The brushes shall conform to the requirements specified herein and shall be similar to the design shown in figure 1. Figure 1 is intended for illustration purposes only and is not intended to preclude brushes otherwise meeting the requirements of this specification. The class 1 brushes shall have the filler material specified in 3.1.1 and the class 2 brushes shall have the filler material specified in 3.1.2. The filler material shall be secured within twisted wires which shall also form the handle of the brush. The hand end of the brush shall have a loop. The filler material shall be approximately parallel to each other and at an angle of approximately 90° to the wire. The filler material shall be evenly distributed in the brush head, shall not separate from the twisted wire when tested as specified in 4.3.1.1, and shall be shaped as shown in the applicable illustrations. The brushes shall meet the dimensional requirements specified in tables I through V. In addition, each style brush shall have the features specified in 3.2.1 and 3.2.3.

3.2.1 Type I - Test tube.

3.2.1.1 Style A dome tip. The distal end of the brush part shall have a dome composed of several fan shaped wings of applicable filler, neatly and securely bound into the twisted wire. In addition, the brush shall meet the dimensional requirements in table I.

TABLE I. Dimensional requirements style A

	Size (Class 1)				Size (Class 2)			
	1/2	3/4	1-1/4	1-1/2	1/2	3/4	1-1/4	1-1/2
Length, over-all minimum inches	8	9-1/2	10-1/4	10-1/4	8	9-1/2	10-1/4	10-1/4
Brush head:								
Length exclusive of dome, minimum inches	2-1/2	3	3-1/4	3-1/4	2-1/2	3	3-1/4	3-1/4
Length dome, minimum inches	1/4	1/2	1/2	1/2	1/4	1/2	1/2	1/2
Diameter, minimum inches	1/2	3/4	1-1/4	1 1/2	1/2	3/4	1-1/4	1-1/2
1/ Filament diamter, minimum inch	-----	-----	-----	-----	.009	.009	.009	.009
Wire (twisted):								
Diameter, single strand minimum inch	.040	.052	.052	.060	.040	.052	.052	.060

1/ H-S-951 tolerances apply.

3.2.1.2 Style B - tuft tip. The distal end of the brush part, classes 1 and 2, shall have a tuft composed of the applicable filler neatly and securely bound into the twisted wire. In addition, the tuft shall be neatly and securely bound near its base with a minimum of three complete turns of 0.016 to 0.018 inch diameter wire normally used to bind the tuft. The brush shall meet the dimensional requirements specified in table II.

TABLE II. Dimensional requirements style B

	Size (Class 1)							Size (Class 2)						
	1/2	3/4	1	1-1/8	1-1/4	1-3/8	1-1/2	1/2	3/4	1	1-1/8	1-1/4	1-3/8	1-1/2
Length, over-all minimum inches	9	9	9	9	9	9	9	9	9	9	9	9	9	9
Brush head: Length, exclusive of tuft min. inches	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Length of tuft minimum inch	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Diameter, minimum inches	1/2	3/4	1	1-1/8	1-1/4	1-3/8	1-1/2	1/2	3/4	1	1-1/8	1-1/4	1-3/8	1-1/2
1/Filament diameter min. inch	---	---	---	---	---	---	---	.009	.009	.009	.009	.009	.009	.009
Wire (twisted): Diameter, single strand, minimum inch	.052	.060	.060	.060	.060	.060	.060	.052	.060	.060	.060	.060	.060	.060

1/ H-S-951 tolerances apply.

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3.2.1.3 Style C - Single fan tip. The distal end of the brush part shall have a single fan shaped wing of applicable filler, neatly and securely bound into the twisted wire. In addition, the brush shall meet the dimensional requirements specified in table III.

TABLE III. Dimensional requirements style C and D

	Style C		Style D	
	Class 1 Size 1/2	Class 2 Size 1/2	Class 1 Size 1-1/2	Class 2 Size 1-1/2
Length, over-all, minimum inches	9	9	13	13
Brush Head:				
Length, minimum inches	2-3/4	2-3/4	2-3/4	2-3/4
Diameter, minimum inches	1/2	1/2	1-1/2	1-1/2
1/ Filament diameter, minimum inch	-----	-----	-----	.009
Wire (twisted):				
Diameter, single strand minimum inch	0.046	0.046	0.0605	0.0605

1/ H-S-951 tolerances apply.

3.2.1.4 Style D - Sponge tip. The distal end of the brush part shall have a piece of synthetic cellulose sponge 7/16 ± 1/32 inch thickness of the commercial quality used for this purpose. The sponge shall be neatly and securely attached to the end of the twisted wire with a minimum of three complete turns of 0.016 to 0.018 inch tie wire normally used to secure the sponge. The sponge shall not pull loose from the end of the brush wire or tear when tested as specified in 4.3.1.2. The brush shall meet the dimensional requirements specified in table III.

3.2.1.5 Style E - Conical. The brush head shall taper from the handle and toward the distal end and shall have a single fan shaped wing of applicable fiber neatly and securely bound into the twisted wire. The diameters and length of taper shall be as specified in table IV.

3.2.1.6 Style F - Conical with tuft tip. The brush head shall taper from the handle end toward the distal end and shall have a tuft, composed of the applicable filler, neatly and securely bound into the twisted wire. In addition, the tuft shall be neatly and securely bound with a minimum of three complete turns of 0.016 to 0.018 inch diameter wire normally used to bind the tuft. The brush shall meet the dimensional requirements specified in table IV.

TABLE IV. Dimensional requirements style E and F

	Style E		Style F			
	Class 1 Size 1/2 - 13/16	Class 2 Size 1/2 - 13/16	Class 1 Size 3/4 - 1-1/2	Class 1 Size 1 - 1-3/4	Class 2 Size 3/4 - 1-1/2	Class 2 Size 1 - 1-3/4
Length, over-all minimum inches	8-1/2	8-1/2	12	9	8-1/2	9
Brush head: Length, minimum excluding tuft, inches	3-1/2	3-1/2	4-1/2	3	4-1/2	3
Diameter, tip minimum inch	1/2	1/2	3/4	1	3/4	1
Diameter, handle end, minimum inches	13/16	13/16	1-1/2	1-3/4	1-1/2	1-3/4
Length of taper, minimum inches	-----	-----	2-1/2	2-1/2	2-1/2	2-1/2
1/ Filament diameter, minimum inch	-----	.009	-----	-----	.009	.009
Wire (twisted): Diameter, single strand, minimum inch	.046	.046	.0605	.0605	.0605	.0605

1/ H-S-951 tolerances apply.

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3.2.2 Type II - Pipet.

3.2.2.1 Style A - Conical. The brush head shall have two (2) diameters with the small diameter at the distal end which shall blend into the larger diameter at the handle end (see illustrations Figure 1). The brush shall meet the dimensional requirements specified in table V.

3.2.2.2 Style B - open tube. The brush head shall be located equidistant from the ends. The handle end of the twisted wire shall have a loop and the other end of the wire shall be tightly twisted to a rounded, smooth end. The brush shall meet the dimensional requirements specified in table V.

TABLE V. Dimensional requirements style A and B

	Style A		Style B	
	Class 1 Size 1/4 - 3/4	Class 2 Size 1/4 - 3/4	Class 1 Size 1/2	Class 2 Size 1/2
Length, over-all minimum inches	17	17	24	24
Brush head:				
Length, minimum inches	7-1/2	7-1/2	3	3
Diameter, minimum inch	-----	-----	1/2	1/2
Minor diameter, minimum inch	1/4	1/4	---	---
Length minor diameter, minimum inches	4	4	---	---
Major diameter, minimum inch	3/4	3/4	---	---
1/ Filament diameter, minimum inch	-----	.008	---	---
Wire (twisted):				
Diameter, single strand minimum inch	.0328	.0328	.0328	.0328

1/ H-S-951 tolerances apply.

3.2.3 Illustration. The illustrations shown herein (figure 1), are for the purpose of identification and are not itended to preclude any brush which otherwise meets the requirements of this specification.

3.3 Sterilization. Hog bristles shall be sterilized at some point in the manufacturing process, after thorough washing and rinsing, in accordance with any one of the procedures specified in 3.3.1, 3.3.2, or 3.3.3. A properly calibrated steam-pressure gauge (applicable for the method in 3.3.3) and a thermometer accurate to ± 2 degrees (for all methods) both visible from the outside of the chambers shall be provided with a suitable recording device to provide a record of the pressure, when applicable, and temperature of the sterilization chamber.

3.3.1 Boiling water. Hog bristle shall be soaked in boiling water (212°F.) for not less than 20 minutes.

3.3.2 Dry heat. Hog bristle shall be placed in a chamber and a temperature of not less than 284°F. shall be developed and maintained for not less than 3 hours.

3.3.3 Autoclaving. Hog bristle shall be placed in a pressure chamber for a minimum of 20 minutes at not less than 15 pounds per square inch at a temperature not less than 248°F.

3.4 Workmanship. The brushes shall be free from defects which affect the appearance, serviceability, and durability. The brushes shall be neatly constructed and finished, and shall have no sharp edges. The filler material shall be free of foreign and loose material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification, or applicable purchase document.

4.2.2 End item examination. The lot shall consist of all brushes of one type, style, class and size offered for inspection at one time. The sample unit shall be one completely fabricated brush.

4.2.2.1 Visual examination. Examination shall be made of the end item for the defects in table VI The inspection level shall be level I and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE VI. Visual defects

Examine	Defects
Material	Not natural unbleached white hog bristle when specified. Not synthetic filament when specified. Brush wire not zinc-coated (glavanized) or corrosion-resistant steel brush wire.
Construction and Workmanship:	
Brush Brush head	Not type, style, class or size specified. Filler materials not evenly distributed; not firmly held in place; can be removed by pulling with thumb and forefinger. Filler material not evenly trimmed or properly shaped (see applicable figure). Not provided with tuft or sponge tip or fan shaped wings at the brush end as specified. Tuft or sponge tip not securely bond with wire. Fan shaped wing or wings not neatly and securely bound into the twisted wire. The synthetic filament not black or the color specified.
Workmanship	Filler material not free of foreign or loose material. Brushes not free of defects that affect their appearance, serviceability or durability.

4.2.2.2 Dimensional examination. Inspection shall be made of the end item for compliance with dimensions in applicable table. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The end item shall be tested as specified in 4.3. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine whether packaging, packing and marking as required by section 5 of this specification are complied with. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping container offered for inspection at one time (see table VII). The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE VII. Examination of preparation for delivery

Examine	Defects
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Contents	Number per container is more or less than specified.
Materials	Any component missing or damaged.

4.3 Tests.

4.3.1 Retention.

4.3.1.1 Filler material. A group of six to twelve individual strands of filler material (bristle or filament) shall be grasped between the thumb and forefinger, and a pulling force shall be applied in a direction tending to pull the strands from the twisted wire. Tests shall be made at three difference places on each brush to determine compliance with 3.2.

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4.3.1.2 Sponge tip. The sponge tip shall be grasped between the thumb and forefinger, and a pulling force shall be applied in a direction tending to pull the sponge from the end of twisted wire to determine compliance with 3.2.1.4.

4.3.1.3 Synthetic filament. Filaments for class 2 brushes shall be tested in accordance with H-S-951 to insure compliance with 3.1.2.

4.3.1.4 Length of bristle or filament. The length of the bristles or filament shall be determined by means of a scale or tape graduated in 1/16 inches or less.

4.3.1.5 Diameter of wire. The diameter of the wire taken from the brushes shall be determined by means of a micrometer caliper graduated to read 0.001 inch or less.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be in accordance with the requirements of PPP-B-1729 as specified for the applicable levels (see 6.2).

5.2 Marking. Containers of the brushes shall be marked in accordance with the requirements of PPP-B-1729.

6. NOTES

6.1 Intended use. The type I brushes are intended for use in cleaning test tubes. The type II brushes are intended for use in cleaning pipets.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, class, and size of brush required (see 1.2.1).
(c) Color of filament for class 2 brushes if other than black is desired (see 3.1.2).
(d) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Comparison of classifications between Federal Specification H-B-173 and H-B-001051(GSA-FSS) covering brushes, test tube and pipet, is as follows:

H-B-173	H-B-001051
Type V, style A	Type I, style A, class 1
None	Type I, style A, class 2
Type V, style B	Type I, style B, class 1
None	Type I, style B, class 2
None	Type I, style C, class 1
None	Type I, style C, class 2
None	Type I, style D, class 1
None	Type I, style D, class 2
None	Type I, style E, class 1
None	Type I, style E, class 2
None	Type I, style F, class 1
None	Type I, style F, class 2
Type IV	Type II, style A, class 1
None	Type II, style A, class 2
Type VI	Type II, style B, class 1
None	Type II, style B, class 2

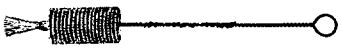
6.4 Supersession. That part of Federal Specification H-B-173 that pertains to type IV, V and VI is superseded by this Federal Specification. The remainder of H-B-173 will remain in effect until Federal Specification H-B-1049 and H-B-1050A are published with appropriate supersession data, at which time a cancellation will be published for H-B-173.

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Type I BRUSHES



Style A



Style B



Style C



Style D

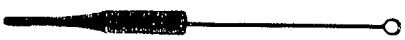


Style E

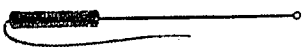


Style F

Type II BRUSHES



Style A



Style B

FIGURE 1

MILITARY INTEREST:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

User activities:

Navy - MC, SA
Air Force - 84
Army - GL

Preparing activity:

GSA

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